MedVQA-Router: Bridging Binary Classification and Open-Ended Captioning for Medical Visual Question Answering

Anonymous Author(s)
Affiliation
Address
email

Abstract

Medical Visual Question Answering (Med-VQA) requires simultaneous understanding of clinical images and natural-language questions. We address both the binary (yes/no) and open-ended subsets of Med-VQA using the `robailleo/medical-vision-llm-dataset` [12], which contains 4,793 samples across X-ray, CT, MRI, and ultrasound modalities. Our two-stage pipeline routes questions via a rule-based classifier, directing binary questions to a multimodal fusion model (ResNet-18 + BioBERT) and open-ended questions to GPT-5 Mini via the OpenAI API. For binary VQA we compare text-only baselines (TF-IDF + LR/SVM), an image-only CNN (ResNet-18), and fusion models. Our best binary classifier—ResNet-18 fused with fine-tuned BioBERT—achieves **79.5% accuracy** and **0.83 AUC-ROC**, outperforming single-modality baselines by over 7 percentage points. For the open-ended branch, we evaluate four vision-language models and find that GPT-5 Mini achieves the highest BERTScore F1 (0.7528) and Recall (0.8021), while InstructBLIP leads on ROUGE, demonstrating that GPT-5 Mini captures clinical meaning more completely than smaller fine-tuned models.

1 Introduction

Medical Visual Question Answering (VQA) has attracted significant research interest as a tool for automated clinical decision support. A Med-VQA system must jointly reason over a clinical image and a natural-language question to produce a correct answer—either binary ("Is there a fracture?") or open-ended ("What abnormality is visible?"), requiring different modelling strategies.

Medical datasets are limited in size due to the high cost of expert annotation and privacy regulations. Clinical images exhibit significant visual diversity across modalities, while questions use a highly specialized vocabulary. Standard vision-language models pretrained on natural images often fail to capture subtle pathological markers critical for accurate diagnosis.

We propose a two-stage pipeline that prioritizes task specialization: a routing classifier separates binary from open-ended questions, directing each to a dedicated model. For binary VQA, a multimodal fusion of ResNet-18 and BioBERT achieves 79.5% accuracy and 0.83 AUC-ROC, outperforming general-purpose models like CLIP. For open-ended captioning, we evaluate a progression of four vision-language models and find that GPT-5 Mini captures clinical meaning most completely, achieving the highest BERTScore F1 (0.7528) and Recall (0.8021).

Submitted to 39th Conference on Neural Information Processing Systems (NeurIPS 2025). Do not distribute.

2 Background

Visual and Medical VQA. Since the VQA v2 dataset [3], VQA research has evolved from monolithic architectures to complex attention-based mechanisms. In the medical domain, datasets such as VQA-Med [5] and SLAKE [9] benchmark performance on radiology and pathology images, where accuracy is safety-critical and linguistic bias is a known challenge.

Transfer Learning and Biomedical Language Models. Due to the scarcity of annotated medical images, transfer learning from ImageNet-pretrained ResNet architectures [4] is standard practice. BERT [2] has spawned specialized variants such as BioBERT [6], pretrained on PubMed abstracts and PubMed Central full texts, making it significantly better suited to clinical terminology than general-domain models.

Vision-Language Models. CLIP [11] aligns images and text in a shared latent space via dual encoders, while BLIP [8] enables flexible generation through a multimodal encoder-decoder mixture. GPT-5 Mini extends this to large-scale instruction-following captioning via the OpenAI API.

3 Data

3.1 Dataset

We use the `robailleo/medical-vision-llm-dataset` [12] from HuggingFace, a combined medical vision-language dataset comprising **4,793 samples** drawn from two sources: ROCO (3,000 samples) and VQA-RAD (1,793 samples). The official split yields **3,834 training** and **959 validation** samples.

Modality distribution: X-ray dominates with 2,325 samples (48.5%), followed by CT (1,351; 28.2%), unknown modality (812; 16.9%), MRI (231; 4.8%), ultrasound (70; 1.5%), and microscopy/endoscopy (4; $<$0.1%). **Body-part distribution:** Unknown (2,856), chest (718), abdomen (468), head (408), extremities (125), spine (119), and pelvis (99).

3.2 Preprocessing and Split

Images are resized to 224×224 and normalized using ImageNet statistics. Grayscale images are converted to three-channel RGB; questions are tokenized to a maximum of 128 tokens. After routing, the binary subset contains 940 samples (750 train / 190 val), nearly balanced at 49.5% / 50.5% yes/no. The open-ended subset contains the remaining 3,853 samples (3,084 train / 769 val), comprising all non-binary answers and drawn primarily from ROCO.

4 Methods

Our system has three components: (1) a routing classifier, (2) a binary VQA pipeline, and (3) an open-ended captioning pipeline.

4.1 Routing Classifier

A rule-based router in `src/utils.py` inspects each answer: "yes" or "no" answers are directed to binary classification; all others to open-ended captioning. The routing is deterministic and error-free by construction, operating on ground-truth answer labels across all 4,793 samples.

4.2 Text-Only Baselines

Questions are vectorized with TF-IDF (10,000 terms, `ngram_range=(1,3)`, `sublinear_tf=True`). We train Logistic Regression (L2, `max_iter=1000`) and SVM (linear kernel) on these features. TF-IDF + SVM achieves 68.95% accuracy and 0.75 AUC-ROC, confirming meaningful but insufficient signal from text alone.

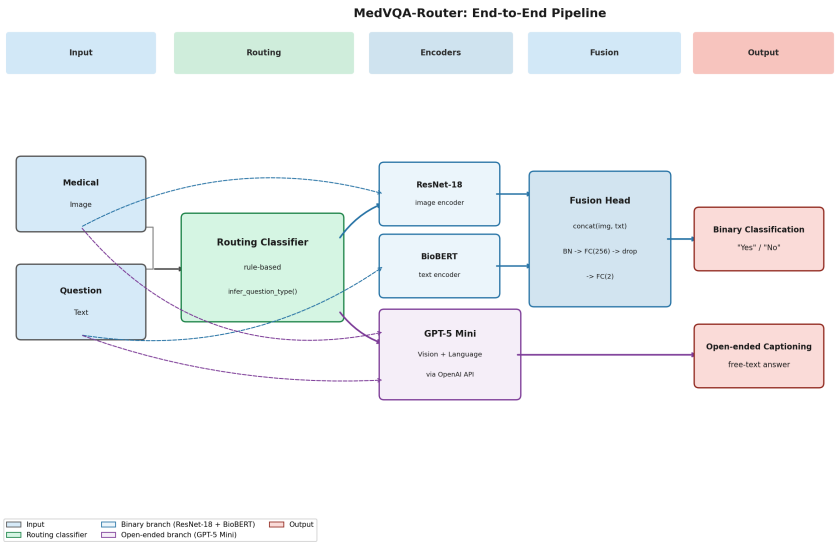

Figure 1: End-to-end pipeline: the routing classifier directs inputs to the binary classification or open-ended captioning branch.

4.3 Image-Only Baseline

ResNet-18 [4] pretrained on ImageNet is fine-tuned with `layer4` unfrozen and a new binary classification head, reusing low-level ImageNet features while adapting high-level representations to the medical domain.

4.4 Multimodal Fusion Models

Our fusion model concatenates image and text features before a learned head:

- **Image encoder**: ResNet-18 (`layer4` unfrozen), 512-dim after global average pooling.
- **Text encoder**:
  Frozen CLIP (512-dim) [11] or BioBERT `dmis-lab/biobert-base-cased-v1.2` [6] (768-dim, mean-pooled last hidden state).
- **Fusion head**: LayerNorm on each stream $\rightarrow$ concatenation $\rightarrow$ Linear(256)–ReLU–Dropout(0.5)–Linear(1).

The fused representation is:

$$z_{\text{fused}} = \left[\, \text{LN}(z_{\text{img}}) \oplus \text{LN}(z_{\text{txt}}) \,\right] \tag{1}$$

where $z_{\text{img}} \in \mathbb{R}^{512}$, $z_{\text{txt}} \in \mathbb{R}^{768}$.

BioBERT is chosen over CLIP because it is pretrained on PubMed abstracts and PubMed Central texts, making it better suited to clinical terminology. We additionally fine-tune BioBERT layers 10–11 at 5×10^{-6} to adapt text representations while avoiding catastrophic forgetting.

Models are trained with `BCEWithLogitsLoss` and AdamW with separate LR groups: 10^{-4} (fusion head), 10^{-5} (ResNet `layer4`), 5×10^{-6} (BioBERT layers 10–11). Batch size 16, weight decay 10^{-4}, up to 30 epochs, early stopping (patience 7), `ReduceLROnPlateau` (factor 0.5, patience 3).

4.5 Open-Ended Captioning Pipeline

We evaluate four vision-language models on the same 769-sample ROCO-derived validation split, using BLIP variants as baselines and GPT-5 Mini as our primary system.

As baselines, we evaluate three BLIP-family models: BLIP [8] (224M parameters, also fine-tuned with frozen vision encoder on 3,084 samples), BLIP-2 [7] (OPT-2.7B backbone, Q-Former bridging

frozen ViT and frozen LLM), and InstructBLIP [1] (Vicuna-7B backbone, instruction-conditioned Q-Former queries). These represent a progression in architectural complexity and serve as reference points for evaluating GPT-5 Mini's capability.

GPT-5 Mini [10] (`gpt-5-mini-2025-08-07`) is queried via the OpenAI API zero-shot, with each image base64-encoded and submitted with the following system prompt:

> *"You are an expert radiologist interpreting medical images. Provide concise, accurate, and specific medical descriptions using precise clinical terminology. Focus only on what is visible in the image."*

Fine-tuning was withheld: the model weights are inaccessible for gradient updates, fine-tuning via API raises reproducibility concerns, and zero-shot performance is the relevant scenario when labeled medical data is scarce. Unlike the BLIP-family models, GPT-5 Mini generates structured multi-sentence clinical reports, a consequence of the radiologist prompt that produces the metric discrepancy discussed in Section 5.3.

Metrics. We report BLEU-4, ROUGE-1, ROUGE-L, and BERTScore [13] (P, R, F1 via DistilBERT). ROUGE measures n-gram overlap and penalises verbosity. BERTScore uses contextual embeddings to capture semantic similarity, correctly recognising that "hepatic metastases" and "liver tumour" are clinically equivalent despite zero lexical overlap.

5 Experiments

5.1 Evaluation Metrics

Binary VQA: accuracy, macro F1, and AUC-ROC on the 959-sample validation split. Open-ended captioning: BLEU-4, ROUGE-L, and BERTScore [13].

5.2 Binary VQA Results

Table 1: Binary VQA model comparison on the validation set.

Model	Accuracy	F1	AUC-ROC
TF-IDF + Logistic Regression	0.6684	0.6441	0.7073
TF-IDF + SVM	0.6895	0.6704	0.7517
ResNet-18 (layer4 unfrozen)	0.7263	0.7500	0.7740
Fusion: ResNet-18 + CLIP	0.7579	0.7723	0.8194
Fusion: ResNet-18 + BioBERT	0.7789	0.7835	0.8204
Fusion: ResNet-18 + BioBERT (fine-tuned)	**0.7947**	**0.8000**	**0.8281**

Results show a clear progression from text-only baselines to multimodal fusion. Moving from ResNet-18 alone (72.63%) to CLIP fusion (+3.16 pp) confirms that visual features alone are insufficient without textual grounding. Replacing CLIP with BioBERT adds a further 2.1 pp, reflecting its stronger clinical vocabulary alignment. Fine-tuning the top two BioBERT layers provides the final boost to 79.47% accuracy and macro-F1 of 0.80.

5.3 Open-Ended Captioning Results

Table 2: Open-ended captioning results on the validation set (769 samples). BLIP-2, InstructBLIP, and GPT-5 Mini are zero-shot. BERTScore uses DistilBERT embeddings. **Bold** = best per column.

Model	BLEU-4	ROUGE-1	ROUGE-L	BS-P	BS-R	BS-F1
BLIP (zero-shot)	0.0000	0.0518	0.0427	0.6539	0.7019	0.6750
BLIP (fine-tuned)	0.0000	0.0953	0.0873	0.6812	0.7025	0.6905
BLIP-2 (zero-shot)	0.0039	0.1409	0.1251	0.7559	0.7304	0.7421
InstructBLIP (zero-shot)	0.0058	**0.1517**	**0.1345**	0.7609	0.7353	0.7467
GPT-5 Mini (zero-shot)	**0.0088**	0.1215	0.0943	0.7107	**0.8021**	**0.7528**

BERTScore F1 improves consistently across all five models (0.675 $\rightarrow$ 0.753), even as ROUGE rankings are less stable. Fine-tuning BLIP's language decoder nearly doubles ROUGE-1 (0.052 to 0.095), confirming domain adaptation helps at small scale. BLIP-2 zero-shot surpasses fine-tuned BLIP on all metrics, showing that a larger language backbone (OPT-2.7B) provides more value than task-specific training with limited data.

The central finding is a rank inversion between ROUGE and BERTScore for GPT-5 Mini. InstructBLIP achieves the highest ROUGE scores (ROUGE-1: 0.152), while GPT-5 Mini leads on BERTScore F1 (0.7528) and Recall (0.8021 vs. 0.735). This occurs because GPT-5 Mini generates verbose structured clinical reports while ROCO references are single-sentence labels—ROUGE penalises every unmatched token, whereas BERTScore captures semantic equivalence in embedding space. GPT-5 Mini's lower Precision (0.711) reflects additional clinical detail beyond the reference, not inaccuracy. This result highlights a methodological limitation: ROUGE underestimates model quality when generated text is richer than ground-truth annotations, making BERTScore the more appropriate primary metric for evaluating frontier VLMs on short-caption datasets.

5.4 Qualitative Analysis

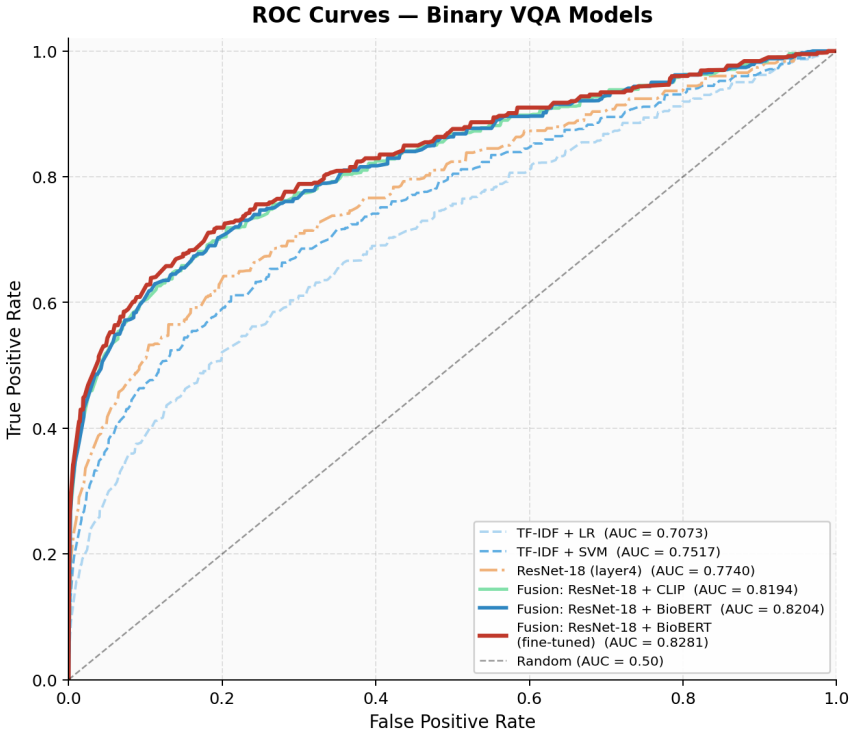

Figure 2: ROC curves for binary VQA models. The fine-tuned BioBERT fusion (AUC = 0.8281) consistently outperforms the vision-only ResNet-18 baseline (AUC = 0.7740).

For the binary branch, the BioBERT stream directed the classification head toward image regions relevant to the specific organ in the query, with failure cases concentrated in images with motion artifacts or fine-grained indicators exceeding the 224×224 resolution. For the open-ended branch, the following example illustrates the ROUGE–BERTScore discrepancy:

> **Reference:** *Computed tomographic scan shows a metastatic liver tumour (arrow).*
>
> **GPT-5 Mini:** *Contrast-enhanced axial CT of the upper abdomen. Numerous well-defined hypoattenuating nodules throughout the hepatic parenchyma (several mm to >1 cm), most consistent with multifocal hepatic metastases. Peripheral/subcapsular lesion along the superior anterior hepatic surface abutting the diaphragm (arrow). Impression: multifocal liver metastases; subcapsular hepatic lesion.*

Same diagnosis, near-zero lexical overlap. ROUGE scores this near zero; BERTScore Recall of 0.802 correctly identifies high semantic coverage.

5.5 Compute Resources

Binary fusion experiments used NVIDIA T4 and A100 GPUs via Google Colab Pro, requiring ≈45 minutes to converge over 30 epochs (batch size 16, max sequence length 128). Open-ended experiments ran on an NVIDIA RTX 5070 Ti (17.1 GB VRAM) locally; BLIP fine-tuning completed in under 2 hours and each zero-shot model took 30–45 minutes. GPT-5 Mini inference over 769 samples took ≈2 hours via the OpenAI API.

6 Conclusion

We presented MedVQA-Router, a two-stage pipeline for Medical VQA that routes questions to a binary classification branch or an open-ended captioning branch. Combining ResNet-18 with fine-tuned BioBERT achieves 79.5% accuracy and 0.83 AUC-ROC, outperforming all single-modality baselines by over 7 pp. The consistent advantage of BioBERT over CLIP confirms the value of domain-adapted language models for medical text understanding. For the open-ended branch, GPT-5 Mini achieves the highest BERTScore F1 (0.7528) and Recall (0.8021) zero-shot, while InstructBLIP leads on ROUGE, revealing that metric choice critically shapes conclusions in medical captioning research. Future work includes cross-attention fusion for finer-grained image-text interaction, end-to-end multi-task routing, and evaluation on larger benchmarks such as VQA-RAD and SLAKE [9].

References

[1] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. InstructBLIP: Towards general-purpose vision-language models with instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[2] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 4171–4186, 2019.

[3] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the V in VQA matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6904–6913, 2017.

[4] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[5] Jason J Lau, Soumya Gayen, Asma Ben Abacha, and Dina Demner-Fushman. A dataset of clinically generated visual questions and answers about radiology images. *Scientific Data*, 5(1):1–10, 2018.

[6] Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, et al. Biobert: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240, 2020.

[7] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International Conference on Machine Learning (ICML)*, 2023.

[8] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International Conference on Machine Learning (ICML)*, pages 12888–12900, 2022.

[9] Bo Liu, Li-Ming Zhan, Li Xu, Lin Ma, Yan Yang, and Xiao-Ming Wu. SLAKE: A semantically-labeled knowledge-enhanced dataset for medical visual question answering. In *IEEE 18th International Symposium on Biomedical Imaging (ISBI)*, pages 1650–1654, 2021.

[10] OpenAI. GPT-5 Mini. `https://openai.com`, 2025.

[11] Alec Radford, Jong Wook Kim, Chris Hallacy, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021.

[12] Robailleo. Medical vision LLM dataset. HuggingFace Datasets. `https://huggingface.co/datasets/robailleo/medical-vision-llm-dataset`, 2024.

[13] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations (ICLR)*, 2020.